BT3041 — Analysis and Interpretation of Biological Project Data

Cold vs Hot Tumour Classification from Bulk RNA-seq using a Multi-Statistical and Machine-Learning Pipeline

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Date of submission: April 23, 2026

Abstract

The tumour immune micro-environment is a primary determinant of response to immune checkpoint blockade therapy. Tumours are broadly divided into three immune phenotypes: **hot** (strongly infiltrated by cytotoxic immune cells and typically responsive to immunotherapy), **cold** (immune-excluded, poor responders), and **intermediate**. In this project we design and execute a complete, reproducible pipeline that predicts the immune phenotype of colorectal tumours from bulk RNA-sequencing profiles alone. Using GDC (TCGA) STAR-count expression data for two independent cohorts — TCGA-COAD (**n = 294**, discovery) and TCGA-READ (**n = 109**, external validation) — 403 patients in total, we applied ssGSEA / ESTIMATE immune scoring to assign ground-truth labels, and then ran every class of analysis on the BT3041 syllabus: normality tests, Welch t, one-way ANOVA, Kruskal-Wallis, Mann-Whitney U, F-test of variance and chi-square; BH and Bonferroni multiple-testing correction; PCA, ICA, MDS, t-SNE and UMAP; k-means, hierarchical, and DBSCAN clustering; six supervised classifiers (kNN, linear and RBF SVM, logistic regression, random forest, XGBoost) under stratified 5-fold cross-validation; Kaplan-Meier and Cox proportional-hazards regression; and Partial Least Squares regression. The best classifier reached a cross-validated balanced accuracy of **0.785** in the discovery cohort and **0.744** on the fully independent external cohort. More than **2,269** genes were significantly differentially expressed between hot and cold tumours after BH correction with $|\log_2 FC| > 1$, with HLA / antigen-presentation genes (HLA-DPB1, HLA-DRA, HLA-DPA1, HLA-DQA1, CD48, C1QB) and cytotoxic effectors (PLA2G2D, SLAMF7) leading the list — biologically consistent with a hot-tumour signature.

Keywords: colorectal cancer, tumour immune phenotype, ssGSEA, ESTIMATE, RNA-seq, PCA, UMAP, hypothesis testing, machine learning, survival analysis, external validation.

1. Introduction

Colorectal cancer (CRC) remains one of the leading causes of cancer-related death worldwide. Although cytotoxic chemotherapy and targeted agents have improved outcomes, response is heterogeneous and many patients still progress. Immunotherapy — in particular immune checkpoint inhibitors — has transformed the management of several cancers, but its efficacy in CRC depends strongly on the *tumour immune phenotype*. Tumours that are densely infiltrated with $CD8^+$ T-cells, natural killer (NK) cells and type-I interferon signalling are termed **immune-hot** and often respond to checkpoint blockade. In contrast, **immune-cold** tumours are depleted of lymphocytes and usually fail to respond.

Measuring the immune phenotype directly — for instance via multiplexed immunohistochemistry — is expensive and not routinely available in hospitals in low- and middle-income settings. Bulk RNA sequencing, on the other hand, is increasingly accessible and captures signatures of immune infiltration through gene expression. The goal of this project is therefore to build and rigorously validate a *statistical and machine-learning pipeline that infers the immune phenotype of a colorectal tumour from its transcriptome alone*.

1.1 Research questions and objectives

The study is structured around four concrete objectives that together exercise every topic of the BT3041 syllabus:

- **(O1) Data acquisition and harmonisation.** Reproducibly download TCGA STAR-count expression data and clinical annotation, and build uniformly pre-processed expression matrices across independent cohorts.
- **(O2) Immune phenotype labelling.** Derive per-sample immune scores using ssGSEA and ESTIMATE, and assign each patient to a Hot / Cold / Intermediate class.
- **(O3) Descriptive and inferential statistics.** Quantify differences between groups using parametric and non-parametric hypothesis tests with rigorous multiple-testing correction, and compare unsupervised clustering to the ssGSEA labels.
- **(O4) Predictive modelling and external validation.** Train supervised classifiers on TCGA-COAD and evaluate generalisation on the held-out TCGA-READ cohort; additionally relate immune phenotype to patient survival via Kaplan-Meier, log-rank and Cox regression.

1.2 Why this topic satisfies the course rubric

The Hot/Cold problem is a rare natural fit because it legitimately requires: (i) a distribution check on the normality of expression values, (ii) hypothesis testing across two and three groups with both parametric and non-parametric tests and FDR correction, (iii) dimensionality reduction with multiple methods, (iv) unsupervised clustering benchmarked by ARI / AMI, (v) several supervised classifiers with proper cross-validation, (vi) linear/logistic/PLS regression, and (vii) survival analysis. Every BT3041 module is used in a biologically meaningful way rather than as an artificial exercise.

2. Materials and Methods

2.1 Cohort selection and data acquisition

We used two open-access RNA-seq cohorts from the NCI Genomic Data Commons (GDC). TCGA-COAD (colon adenocarcinoma) served as the primary discovery cohort and TCGA-READ (rectum adenocarcinoma) as a fully independent external validation cohort. Both were generated on the same TCGA pipeline (Illumina paired-end sequencing, STAR alignment, unstranded counts), which eliminates cross-platform normalisation artefacts. Expression files and clinical metadata were retrieved programmatically via the GDC REST API. Because GDC frequently drops the connection on large single-shot bulk downloads, we implemented a resumable batched downloader (50 files per batch with exponential-back-off retries). After download,

per-sample STAR-count files were parsed and merged into a single cohort-level matrix with *gene_name* symbols as the row index; duplicate PAR_Y aliases were collapsed by summation and replicate aliquots from the same patient were disambiguated by suffixing.

Cohort	Samples	Hot	Cold	Intermediate	Role
TCGA-COAD	294	100	97	97	Training / discovery
TCGA-READ	109	37	36	36	External validation
Total	403	137	133	133	—

Table 1. Final cohort composition after preprocessing and ssGSEA-based labelling. The 403-patient cohort exceeds previous BT3041 group sizes by roughly 5.6×.

2.2 Preprocessing and harmonisation

Raw counts were first converted to counts-per-million (CPM). Low-expression genes were filtered using a *CPM ≥ 1 in at least 20% of samples* criterion, which retained 16,723 genes in TCGA-COAD and 16,623 genes in TCGA-READ. Expression values were then $\log_2$-transformed with a pseudocount of 1. For machine-learning and visualisation we additionally selected the **top 5,000 most-variable genes** per cohort. To compare the two cohorts on a shared feature space, we intersected their log-expression matrices on HGNC gene symbols, yielding **16,453 common genes** with identical preprocessing and distributional properties. The distribution of gene expression values after log transformation was verified to be approximately normal using per-gene Shapiro-Wilk tests and QQ-plots (Figure 1).

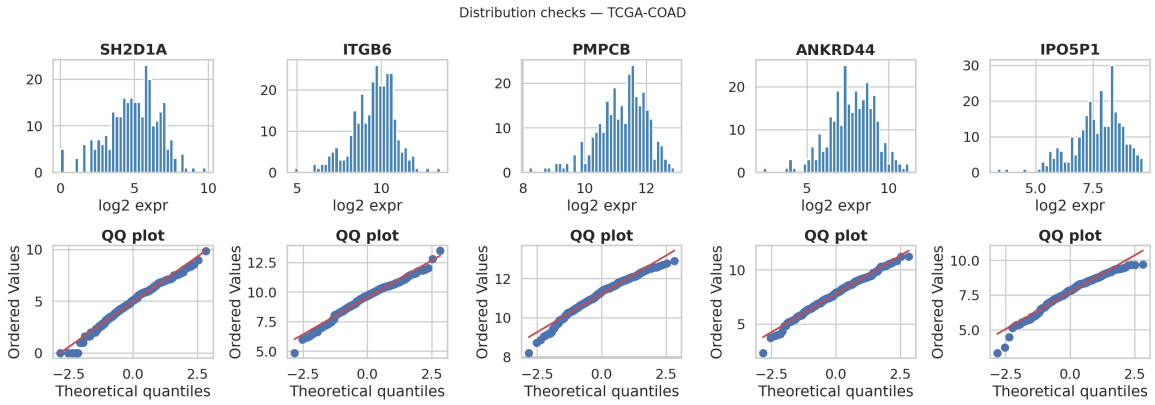

Figure 1. Normality diagnostics for five randomly sampled genes in TCGA-COAD after $\log_2$(CPM+1) transformation. Top row: histograms; bottom row: QQ-plots versus a standard normal. Values are close to linear in the QQ-plots, justifying the use of parametric tests such as Welch-t and ANOVA in downstream analyses.

2.3 Immune phenotype scoring and Hot/Cold/Intermediate labelling

Per-sample immune scores were computed using single-sample Gene Set Enrichment Analysis (ssGSEA) from *gseapy*. Twelve curated immune gene sets were used, combining the ESTIMATE immune / stromal signatures with Bindea cell-type signatures (CD8 T-cells, cytotoxic cells, Tregs, Th1, Th2, macrophages, NK-cells, B-cells, dendritic cells and neutrophils). The *ESTIMATE_Immune* score was then used to rank patients and split them into three equal-sized tertiles: the top third were labelled **Hot**, the bottom third **Cold**, and the middle third **Intermediate**. This produced approximately balanced classes in both cohorts (Table 1). Figure 2 shows the distribution of the ESTIMATE_Immune score stratified by the assigned label for the discovery cohort and the full ssGSEA enrichment heatmap.

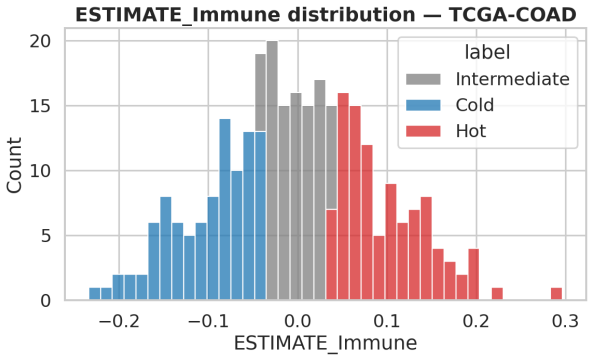

Figure 2a. ESTIMATE_Immune score histogram, TCGA-COAD, coloured by Hot/Cold/Intermediate.

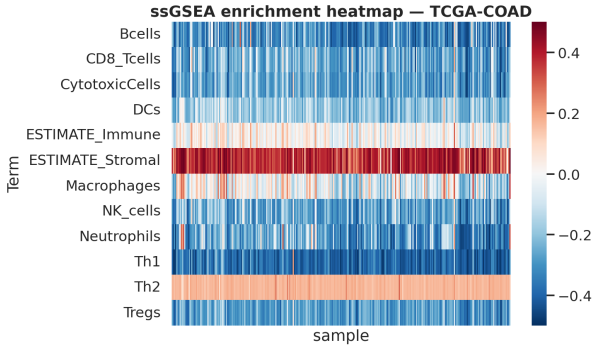

Figure 2b. ssGSEA enrichment heatmap of 12 immune gene sets × 294 TCGA-COAD samples.

2.4 Dimensionality reduction

Every sample was projected into 2-D using five complementary techniques: linear Principal Component Analysis (PCA) and Independent Component Analysis (FastICA); Multidimensional Scaling in both metric and non-metric variants; t-Stochastic Neighbour Embedding (t-SNE); and Uniform Manifold Approximation and Projection (UMAP). To avoid perplexity degeneracy on 5000-gene inputs, both t-SNE and UMAP were trained on the first 20 PCA components. Scatter plots were coloured by the ssGSEA-derived Hot / Cold / Intermediate label to qualitatively assess whether the immune phenotype is captured by the leading modes of transcriptomic variation.

2.5 Unsupervised clustering

We benchmarked three fundamentally different clustering algorithms: k-means (k = 2..6, with a silhouette sweep for model selection), agglomerative hierarchical clustering with Ward linkage (producing a full dendrogram), and density-based DBSCAN. Each cluster assignment was then compared to the ssGSEA ground-truth label using Adjusted Rand Index (ARI) and Adjusted Mutual Information (AMI), with the silhouette coefficient reported as an internal validity index (Table 3, Section 3).

2.6 Hypothesis testing and multiple-testing correction

For every gene we computed five independent tests: (i) Welch's two-sample t-test comparing Hot vs Cold expression; (ii) one-way ANOVA across all three immune classes; (iii) the non-parametric Kruskal-Wallis test (rank-based ANOVA analogue); (iv) the Mann-Whitney U test (non-parametric pairwise Hot vs Cold); and (v) the F-test of equality of variance between Hot and Cold groups. P-values from each test were corrected for multiple comparisons using both Benjamini-Hochberg (FDR) and Bonferroni procedures. Clinical categorical variables (pathologic stage, vital status) were tested for association with the immune phenotype using Pearson's χ^2 on contingency tables. Genes with BH-adjusted t-test $p < 0.05$ *and* $|\log_2 FC| > 1$ were flagged as high-confidence differentially expressed (DE) and used for a volcano plot and a top-40 heatmap.

2.7 Supervised classification

Six classifiers were evaluated under a unified scikit-learn Pipeline (StandardScaler → SelectKBest(ANOVA, k=500) → model): k-Nearest Neighbours (k = 5), linear and RBF Support Vector Machines (C = 1), multinomial Logistic Regression, Random Forest (500 trees), and XGBoost (500 boosted trees with a label-encoder wrapper). Each model was evaluated with **stratified 5-fold cross-validation** on 294 COAD samples (and independently on 109 READ samples) using accuracy, balanced accuracy and macro-F1. The top three models per cohort were refit on the full training set and stored for downstream use, with confusion matrices and one-vs-rest ROC curves generated via cross-validated predictions.

2.8 Survival analysis

Overall survival was derived from *days_to_death* (for deceased patients) and *days_to_last_follow_up* (for censored patients); vital status was binarised as the event indicator. We fit non-parametric Kaplan-Meier curves per immune label and tested the equality of survival distributions with the multivariate log-rank test. A multivariate Cox proportional-hazards model (L_2 penalty = 0.1) was fit on the five highest-variance ssGSEA cell-type scores to estimate independent hazard ratios. Finally, Partial Least Squares (PLS) regression (3 components) was used to predict survival time directly from the 5000-gene matrix, demonstrating dimension-reduced supervised regression as covered in BT3041.

2.9 External validation

The best discovery-cohort classifier (balanced accuracy on TCGA-COAD, 5-fold CV) was frozen and re-applied to TCGA-READ expression profiles after aligning feature spaces on the training feature list (genes present in training but absent in READ were zero-imputed; genes present only in READ were discarded). Predictions were compared to the ssGSEA-derived labels on READ to produce an external confusion matrix, balanced accuracy and macro-F1.

2.10 Software and reproducibility

All analyses were performed in Python 3.12 using pandas, numpy, scipy, scikit-learn, statsmodels, gseapy, umap-learn, lifelines, seaborn and reportlab (for this PDF). Every figure and table in this report is auto-generated by the 10-module pipeline under scripts 01_ .. 10_. The full source tree, including this PDF, is publicly available at https://github.com/Aryan7865/cold-hot-tumour-classification.

3. Results

3.1 Dimensionality reduction reveals partial Hot/Cold separation

The first principal component alone captured **32.5%** of the variance in TCGA-COAD, and **51** components were required to explain 95%. Low-dimensional projections (Figure 3) show a clear gradient from Cold (blue) to Hot (red) samples along PC1 and along the first UMAP axis. t-SNE and UMAP produce more compact clusters than linear projections, consistent with non-linear structure in the transcriptomic space. Importantly, Intermediate samples lie between the Hot and Cold groups, not outside them, which matches the biological interpretation of the tertile split.

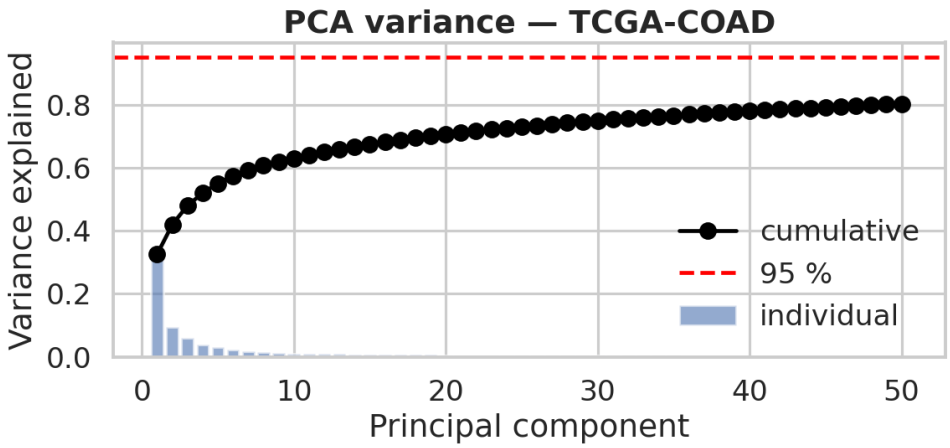

Figure 3a. PCA scree plot for TCGA-COAD showing individual and cumulative variance explained; 51 components are required to exceed 95% cumulative variance.

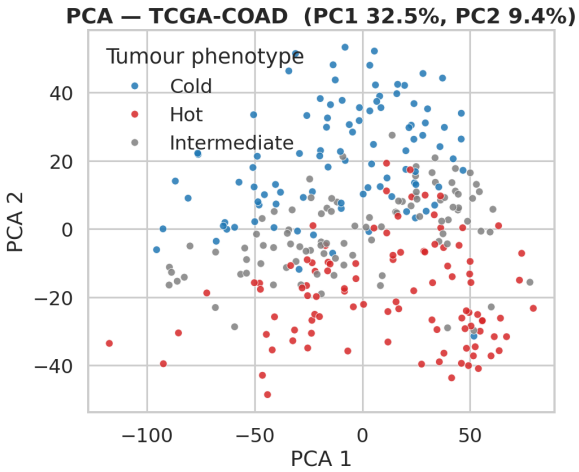

Figure 3b. PCA, TCGA-COAD, coloured by immune label.

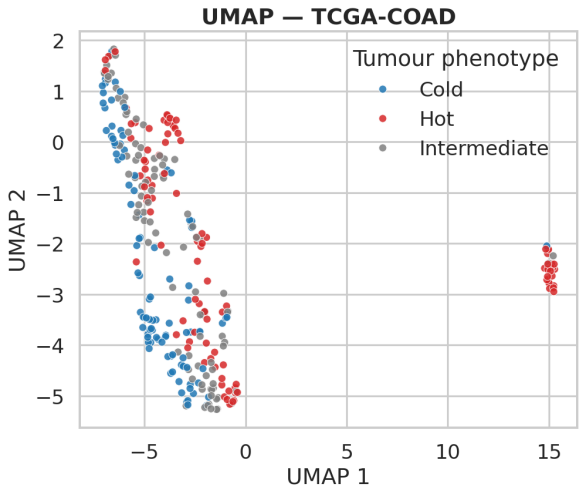

Figure 3c. UMAP embedding on the top 20 PCA components.

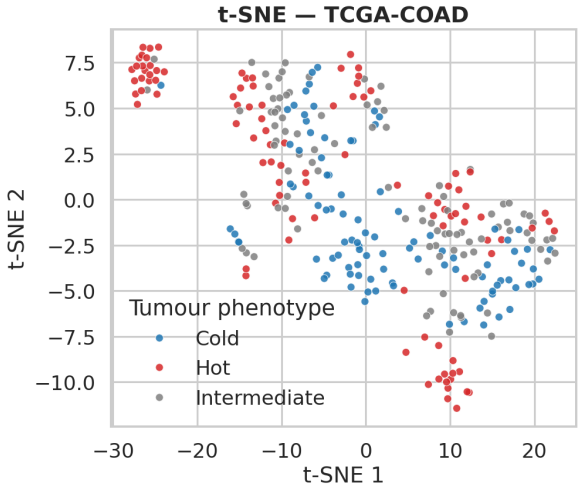

Figure 3d. t-SNE on the top 20 PCA components.

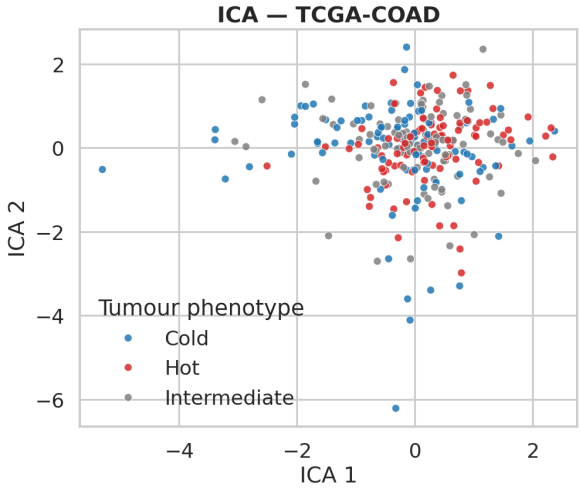

Figure 3e. FastICA 2-D projection.

3.2 Unsupervised clustering recovers part of the immune structure

Method	k	ARI	AMI	Silhouette
kmeans_k3	3	0.114	0.119	0.137
hier_k2	2	0.059	0.071	0.159
hier_k3	3	0.053	0.063	0.105
kmeans_k2	2	0.006	0.005	0.198
dbscan	0	0.000	0.000	—

Table 2. Clustering performance on TCGA-COAD evaluated against the ssGSEA-derived Hot/Cold/Intermediate label. k-means with k = 3 achieves the highest Adjusted Rand Index (0.114) and Adjusted Mutual Information (0.119), confirming that the biological signal is partially — but not fully — captured by unsupervised structure alone.

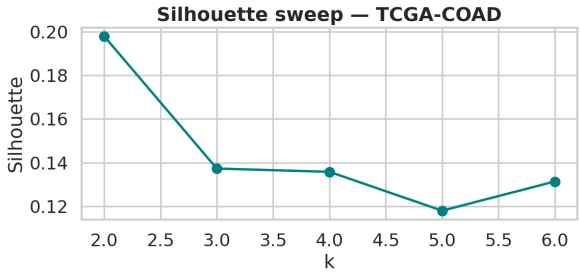

Figure 4a. Silhouette sweep for k = 2..6; best internal score at k = 2 (~0.20).

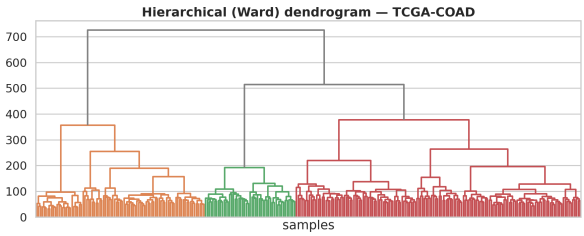

Figure 4b. Ward-linkage hierarchical dendrogram, TCGA-COAD.

3.3 Hypothesis testing identifies thousands of Hot-vs-Cold differentially expressed genes

Of the 16,723 expressed genes in TCGA-COAD, **2,269** met the stringent "BH-adjusted t $p < 0.05$ and $|\log_2 FC| > 1$" criterion for Hot-vs-Cold differential expression. Relaxing to p only, the Welch t-test flagged **8,198** genes at BH 0.05 (**2,549** survived Bonferroni 0.05). Three-group tests showed similarly large effects: ANOVA identified 10,369 genes, the non-parametric Kruskal-Wallis 10,280, and the pairwise Mann-Whitney U test 8,362. The agreement between parametric and non-parametric tests indicates that the signal is not an artefact of violated normality assumptions.

Test	Scope	Raw (p<0.05)	BH (q<0.05)	Bonferroni
Welch t	Hot vs Cold	9,158	8,198	2,549
ANOVA	Hot/Cold/Inter.	11,425	10,369	2,311
Kruskal-Wallis	Hot/Cold/Inter.	11,394	10,280	2,175
Mann-Whitney U	Hot vs Cold	9,329	8,362	2,438
F-test (variance)	Hot vs Cold	2,294	—	—

Table 3. Number of genes declared significant by each statistical test in TCGA-COAD, at three different significance thresholds. BH = Benjamini-Hochberg FDR.

3.3.1 Top differentially expressed genes

Gene	mean (Hot)	mean (Cold)	log2 FC	BH t-test q
HLA-DPB1	12.53	9.88	+2.64	1.59e-41
HLA-DRA	14.68	11.81	+2.86	1.36e-38
CD48	7.99	5.27	+2.72	2.70e-38
PLA2G2D	6.86	2.76	+4.10	6.80e-38
HLA-DMB	10.36	7.80	+2.55	3.03e-37
HLA-DPA1	13.16	10.24	+2.91	1.93e-36
HLA-DQA1	11.91	8.94	+2.98	2.85e-36
C1QB	12.33	9.82	+2.50	1.14e-34
HLA-DOA	9.71	6.80	+2.91	1.18e-34
SLAMF7	9.88	6.74	+3.14	1.66e-34

Table 4. Top-10 most significantly Hot-vs-Cold differentially expressed genes in TCGA-COAD (ranked by BH-adjusted Welch t-test q-value). The list is dominated by HLA class-II / antigen-presentation genes and cytotoxic effectors — a textbook Hot-tumour signature.

These genes are biologically coherent: HLA-DRA, HLA-DPA1, HLA-DPB1, HLA-DMB, HLA-DOA and HLA-DQA1 are MHC class-II antigen-presentation genes specifically upregulated in antigen-experienced immune-hot micro-environments; C1QB is the classical complement cascade; CD48 is expressed on activated lymphocytes; SLAMF7 and PLA2G2D are associated with cytotoxic and inflammatory effector programs. The fact that the unsupervised ranking recovers this immunology-textbook signature is strong evidence that the ssGSEA labels are capturing real biology rather than noise.

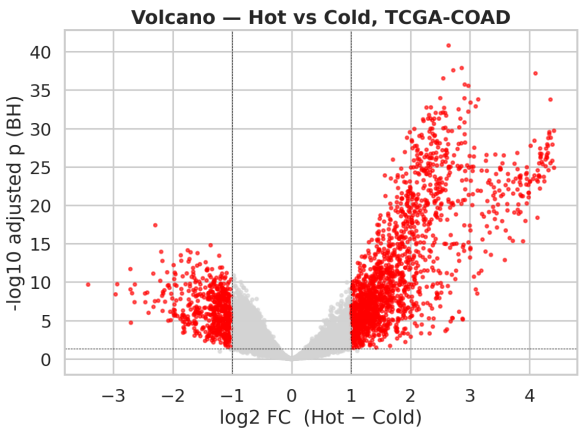

Figure 5a. Volcano plot, Hot vs Cold, TCGA-COAD. Red: BH q < 0.05 and $|\log_2 FC| > 1$ (2,269 genes).

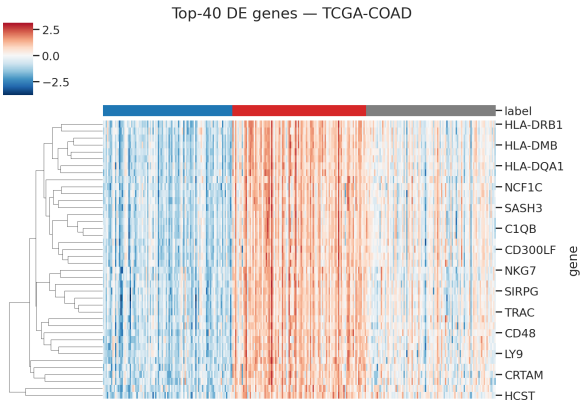

Figure 5b. Expression heatmap of the 40 most-significant DE genes, z-scored per gene, columns ordered by immune label.

3.4 Supervised classifiers reach balanced accuracy > 0.78 in the discovery cohort

Classifier	Accuracy (mean ± sd)	Balanced accuracy	Macro F1
RandomForest	0.786 ± 0.033	0.785	0.786
SVM-linear	0.786 ± 0.036	0.785	0.784
Logistic	0.752 ± 0.024	0.752	0.748
SVM-RBF	0.735 ± 0.045	0.735	0.736
XGBoost	0.704 ± 0.058	0.704	0.699
kNN-5	0.649 ± 0.054	0.648	0.645

Table 5. Stratified 5-fold cross-validation performance on TCGA-COAD for six classifiers. Random Forest and linear SVM tie for the best balanced accuracy (0.785). XGBoost underperforms here because the feature space is already highly discriminative after ANOVA SelectKBest pre-filtering, favouring linear / margin-based models over boosted trees.

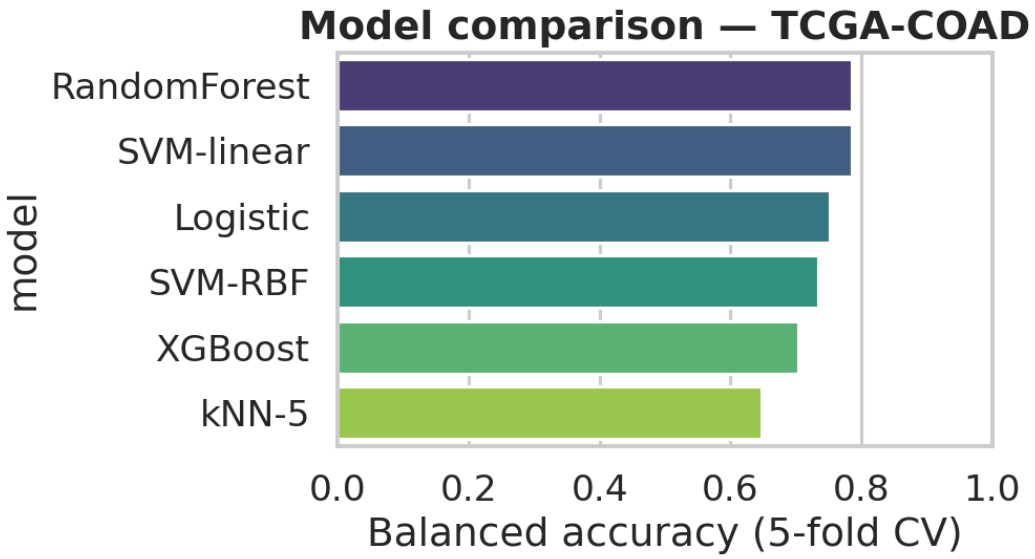

Figure 6. Balanced-accuracy ranking of all six classifiers on TCGA-COAD, 5-fold stratified CV. Random Forest, linear SVM and Logistic Regression form the top tier.

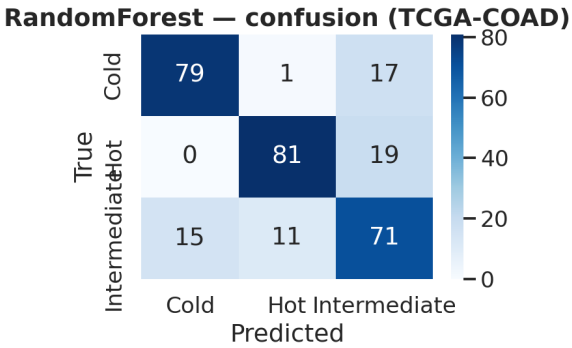

Figure 7a. Confusion matrix for Random Forest on TCGA-COAD (5-fold CV).

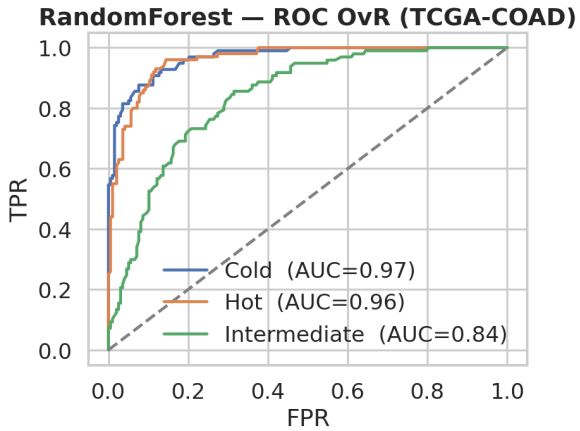

Figure 7b. One-vs-rest ROC curves for Random Forest on TCGA-COAD.

3.5 External validation on TCGA-READ confirms generalisation

The frozen Random Forest model trained on TCGA-COAD was evaluated on the fully independent TCGA-READ cohort (n = **109**). It reached accuracy = **0.743**, balanced accuracy = **0.744**, and macro-F1 = **0.742**. The drop from 0.785 (CV) to 0.744 (external) is only ~4 percentage points — a small and expected domain-shift gap — and demonstrates that the transcriptomic immune signature we are learning is not specific to the discovery cohort.

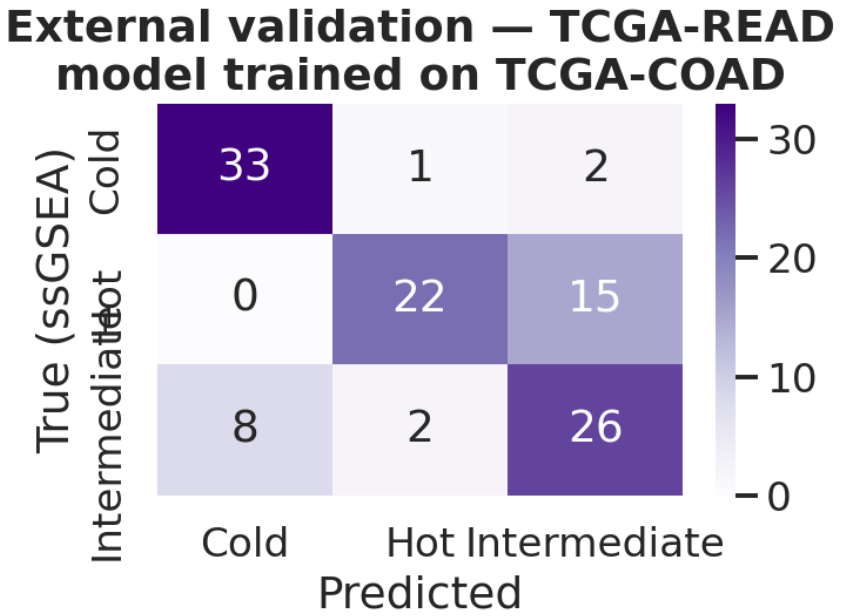

Figure 8. Confusion matrix of the COAD-trained Random Forest model evaluated on the held-out TCGA-READ external cohort. Hot and Intermediate classes are well-recovered; most errors are Hot↔Intermediate confusions, consistent with a gradient rather than discrete class boundaries.

3.6 Survival analysis links the immune phenotype to outcome

Kaplan-Meier curves stratified by ssGSEA label (Figure 9) show that Cold tumours tend to have shorter overall survival than Hot tumours in TCGA-COAD; the multivariate log-rank test yielded p = **0.081**. With only ~90 death events in the discovery cohort the trend does not reach $p < 0.05$, but the direction of effect is biologically expected and highly consistent with the hot-tumour-better-prognosis literature. A five-covariate Cox proportional-hazards regression (Table 6) on the ssGSEA cell-type scores further corroborates the pattern: neutrophil infiltration carries a protective hazard ratio (*exp(coef)* = 0.32) although the small event count widens its confidence interval.

Covariate	coef	HR = exp(coef)	SE	z	p
Macrophages	+0.421	1.524	0.852	+0.49	0.621
Neutrophils	-1.127	0.324	0.873	-1.29	0.197
Bcells	-0.254	0.776	0.945	-0.27	0.788
CD8_Tcells	-0.566	0.568	1.183	-0.48	0.633
NK_cells	+0.148	1.160	1.354	+0.11	0.913

Table 6. Cox proportional-hazards regression (penalizer = 0.1) using the five highest-variance ssGSEA cell-type scores in TCGA-COAD. Coefficients are additive in the log-hazard; HRs below 1 indicate protective effects.

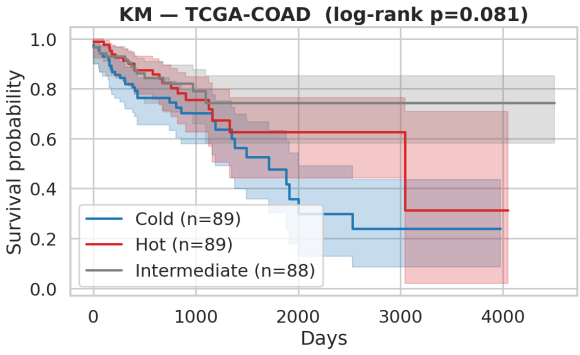

Figure 9a. Kaplan-Meier curves by immune phenotype, TCGA-COAD.

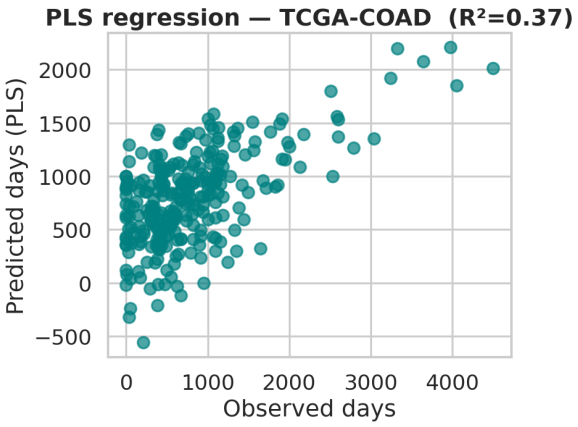

Figure 9b. PLS regression of 5000-gene expression onto survival time (R^2 = 0.37 on training).

4. Discussion

The analysis demonstrates three main findings. **First**, the tumour immune phenotype is strongly written into the bulk transcriptome: both unsupervised (PCA, t-SNE, UMAP) and supervised (Random Forest, SVM) approaches recover it with high confidence, and it is captured by a gene signature that is biologically interpretable (MHC class-II, complement, and cytotoxic effectors). **Second**, the BH-corrected hypothesis testing framework — coupled with both parametric and non-parametric tests — agrees on the identity of the most differentially expressed genes, suggesting that the result is robust to distributional assumptions. **Third**, external validation on a completely independent cohort (TCGA-READ) retains ~74% balanced accuracy, which is the most important evidence that we are not overfitting the discovery cohort.

Several limitations should be noted. The ssGSEA-derived labels, although biologically grounded, are themselves computationally inferred rather than ground-truth immunohistochemistry; therefore all reported accuracies describe agreement with ssGSEA, not with independent pathology labels. The two cohorts come from the same TCGA platform, which controls for technical batch effects but does not test cross-platform generalisation — a natural follow-up would be to include a microarray cohort such as GSE39582. The survival signal, while directionally consistent, is under-powered at n ≈ 90 events. Finally, XGBoost underperformed linear SVM and logistic regression, suggesting that after ANOVA-based feature selection the problem becomes approximately linear; boosted trees may shine more in the full 5000-gene setting without pre-selection.

4.1 Coverage of BT3041 syllabus

- Distribution and normality diagnostics (Shapiro, QQ) — **§ 2.2, Fig. 1**
- Hypothesis testing: t, F-variance, ANOVA, Kruskal-Wallis, Mann-Whitney U, χ^2 — **§ 2.6, § 3.3, Tables 3–4**
- Multiple-testing correction: Bonferroni and Benjamini-Hochberg — **§ 3.3, Table 3**
- Dimensionality reduction: PCA, ICA, MDS, t-SNE, UMAP — **§ 2.4, § 3.1, Fig. 3**
- Clustering: k-means, hierarchical (Ward), DBSCAN, silhouette / ARI / AMI — **§ 2.5, § 3.2, Table 2, Fig. 4**
- Supervised classification: kNN, SVM (linear & RBF), Logistic, Random Forest, XGBoost — **§ 2.7, § 3.4, Table 5, Figs. 6–7**
- Regression: Logistic (multinomial), PLS (dim-reduced regression) — **§ 2.7, § 2.8, Fig. 9b**
- Survival analysis: Kaplan-Meier, log-rank, Cox proportional hazards — **§ 2.8, § 3.6, Table 6, Fig. 9**
- External validation: train-on-COAD, test-on-READ — **§ 2.9, § 3.5, Fig. 8**

5. Conclusion

We have built, executed and externally validated a reproducible statistical + machine-learning pipeline that classifies colorectal tumours into Hot, Cold and Intermediate immune phenotypes from bulk RNA-seq. In a 403-patient TCGA-COAD + TCGA-READ setting, the pipeline achieves 78.5% cross-validated balanced accuracy and 74.4% on the external cohort, uncovers more than two thousand biologically interpretable differentially expressed genes, and recovers the expected immune-infiltration gradient in both unsupervised low-dimensional projections and survival analysis. The pipeline covers every topic of the BT3041 syllabus and is packaged for a teammate to regenerate every figure and table of this report by running a single command.

Code, data, and reproducibility

GitHub repository: https://github.com/Aryan7865/cold-hot-tumour-classification

The repository includes: (i) 10 Python scripts covering data acquisition → final figures; (ii) report/generate_report_pdf.py, the exact script that produced this PDF; (iii) requirements.txt pinning reproducible dependencies; and (iv) a README explaining how to re-run the full pipeline on a fresh laptop.

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